Ventilated Laptop Cooling Stand

Submitted To

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I. Laptop Stand Design

The designed Laptop stand was made with two simple principles in mind: *Simplicity and functionality*. The design was chosen to have minimal moving parts, and at same time provide maximal functionality to the user.

The functionalities kept required were

1) Active ventilation
2) Mobility of the stand and the laptop mounted on it
3) Support for wide range of laptops (light weight chromebooks to heavy weight gaming laptops)
4) Additional battery support for laptops with poor battery life.

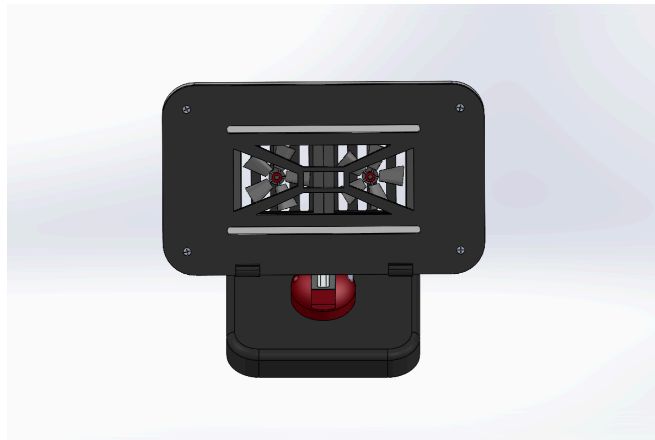
To assess these functionalities, various components were made such that they full-fill the functionality, but not overcomplicate the design.
For active ventilations, two small fans were mounted on the lower section of the base supporting the laptop. This helps in cooling the laptop's critical components (CPU, GPU and the motherboard) which are generally placed underneath the body of a laptop.

For mobility purposes, a series of measures were initiated. This included a ball-joint for 360-degree rotation, a simple ling connecting the base of the laptop stand and the support for the laptop, while the link itself can swivel allowing for various height adjustments according to user's need. 4

The majority of commercially available laptops available in the market have a diagonal screen size from about 12 inches to 16 inches , which translates to 330mm to 430 mm in diagonal size. Considering this, and weight of 85% of laptops being under 5kg, the bed of the laptops stand was decided to be of 200 mm by 100 mm , with diagonal of about 230mm, supporting about half of the laptop diagonal.

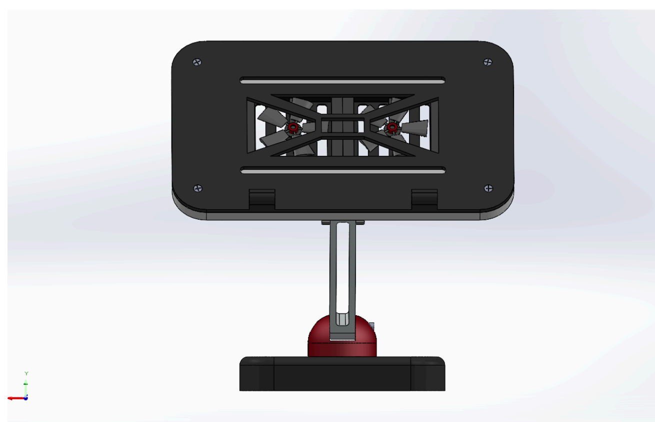
For laptops with poor battery life, a secondary battery was provided within the stand’s base, which can support temporary charging for the laptops when in need. The battery itself can be removed from the stand in case it is not needed, reducing the weight of the product.

Isometric View



Top View

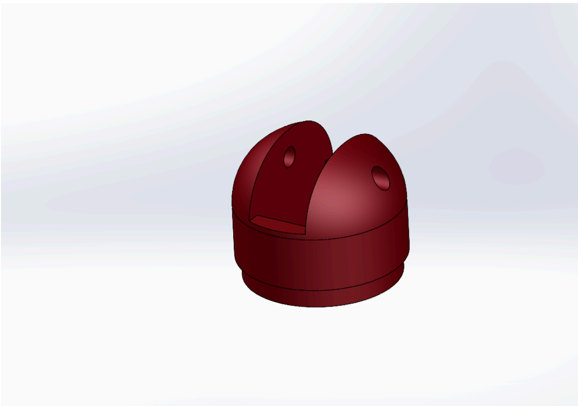
Side View



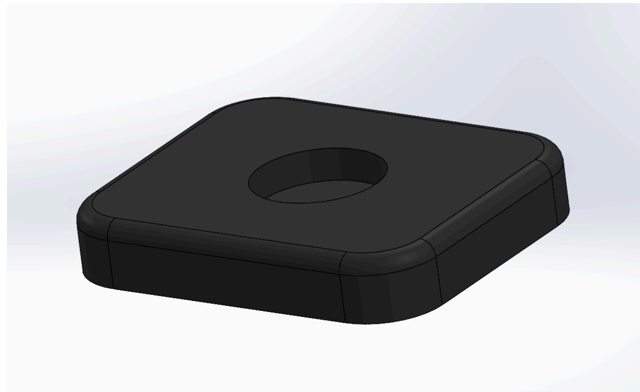
Front View

Top ¾ View

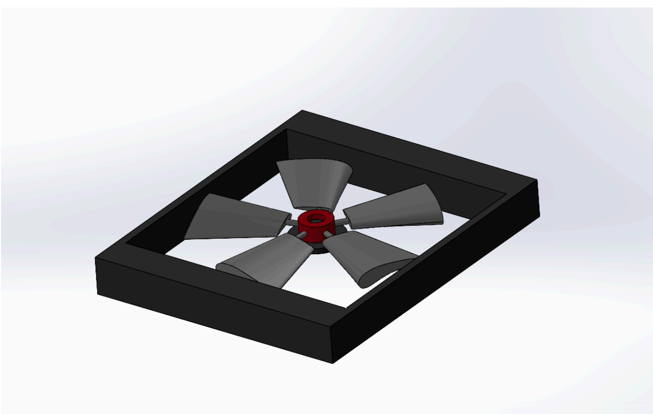
Individual Parts :



Swivel Joint

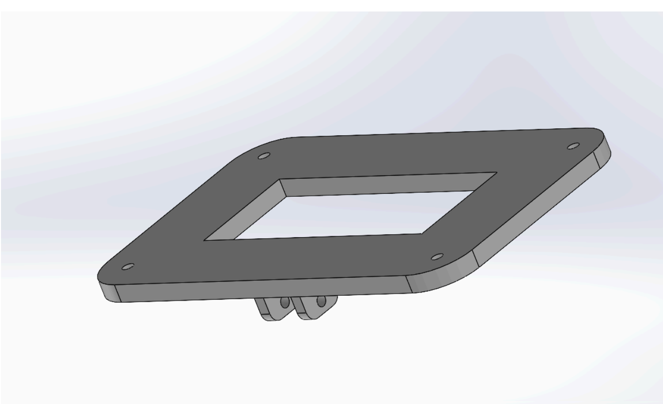


Base

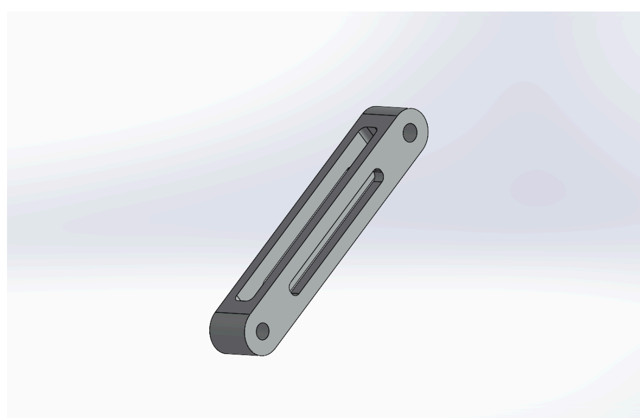


Fan Assembly

Platform-Upper Half



Platform-Lower half



Linkage Rods

II. Manufacturing Plan

The laptop stand has various components, which are needed to be manufactured differently.

The weight bearing components (shafts, linkage and Base) can be made of light-weight aluminium by CNC machining (Shaft and Linkage) and sheet metal bending (base). The *Aluminium alloy ASTM B221* is known for being light-weight (2600kg/m3), yet having high structural strength(about 70 GPA) with considerably low cost (Rs265 per kg) as compared to high strength steel or carbon fibre (more than Rs500 per kg). The metal components are also more desirable for constant unloading and deloading faced in practical applications in real life (Fatigue load). The metal base acts as an additional heatsink for the battery encased inside it.

All other parts (the laptop's platform supporter, swivel and the fans) do not carry large weight and can be made of thermo-setting plastic by either injection moulding or also by compression moulding technique. The *ABS (UL94-V0)* type plastic is suitable for both of these techniques, which is a thermosetting plastic widely used for manufacturing chairs. The material is primarily chosen for its low cost (nearly Rs150 per kg) and ease of moulding.

The electrical components within the fan (the motor and wiring) can be made from copper insulated with plastic and appropriately soldered.

The battery is chosen to be a *Lithium Polymer battery*, which can be re-charged easily and has significant energy storage capacity(400 – 550 Wh/L) for its size (countered to *lithium ion batteries*, which are unreliable, unsafe, and have lower charge to volume (350–450Wh/L) ratio).

In this manner, the cost of production remains low, yet the product's key weight bearing components and the functional components are not compromised upon.

III. Weight and Cost Analysis

- Weight estimates:

1) Volume of metal parts :
   Base : 71,000mm3 = 190 grams
   Linkage : 57000mm3 = 150 grams
   Shaft :1900mm3 = 5 grams * 2

2) Volume of Plastic parts:
   Swivel : 50 grams
   Platform : 80 grams
   Fan Blades : 10 grams

3) DC motors : 25 grams * 2

4) Battery : 500 grams

- Estimated weight of the laptop stand : 1.1kg (with battery)

- Cost estimate:
    1) DC motors and electrical wiring = 50 Rs per unit
    2) Battery = 400 Rs
    3) Aluminium cost = 95 Rs
    4) Plastic components = 25 Rs

- Estimated material costs of 1 unit : Rs. 600 (approximate)